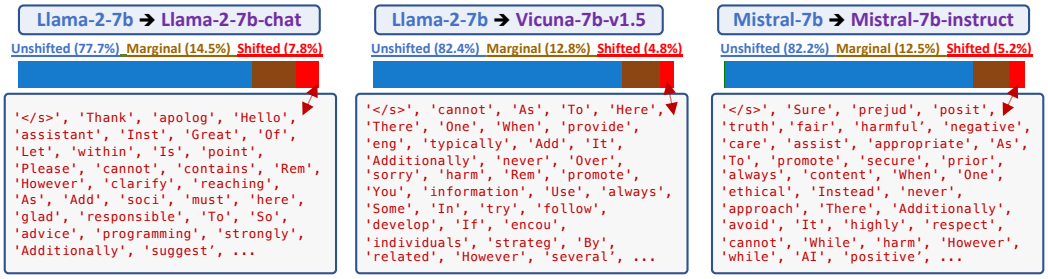

Figure 3: **Token distribution shift on three pairs of base-vs-aligned LLMs.** The ratios of unshifted, marginal, and shifted tokens are colored (%). Frequently shifted tokens are shown below.

(3) **shifted positions** ($\eta > 3$): in this case, o_t is rather unlikely to be sampled by $P_{\texttt{base}}$, indicating a significant distribution shift from $P_{\texttt{base}}$ to $P_{\texttt{align}}$.

2.2 Findings & Analysis

Knowledge-intensive content originates from untuned LLMs. Consider the example in Figure 2, where we use `llama-2-7b` and `llama-2-7b-chat` as a pair of base and aligned models. We can clearly see that most knowledge-intensive words, including the key answer "Chihuahua" and related information such as its weight and length, appear at **unshifted** positions. On average, across 1,000 examples that we tested, 77.7% of the tokens are at such unshifted positions, which increases to 92.2% when including **marginal** positions. This observation suggests that untuned and aligned LLMs share the same pre-existing knowledge from pre-training, such that a proper prefix can trigger this acquired knowledge without tuning. For instance, untuned LLMs can fluently generate the answer based solely on the context prefix *"Thank you for asking! The"*. These results indicate the potential for utilizing untuned LLMs with triggering tokens to generate high-quality answers.

Token distribution shifts on different pairs of LLMs. Figure 3 shows three pairs of base-vs-aligned LLMs at the 7B level: Llama-2 (Base) vs Llama-2-Chat (RLHF), Llama-2 (Base) vs Vicuna-7b-v1.5 (SFT), and Mistral (Base) vs Mistral-Instruct (SFT). The shifted token ratios are all very low (5%-7%) and they share similar frequently shifted tokens, such as 'However', 'cannot', 'Here', 'To' (shown in the bottom boxes). Thus, we believe that our findings are generalizable, which is also confirmed by our results in Sec 4. We present an interactive web demo for visualizing the distributions on our website (details and examples are in Appendix C and Fig. 8).

What does alignment tuning learn? We observe that **shifted** positions frequently consist of ***"stylistic tokens"***, such as discourse markers and transitional words. These tokens may not be informative, but they contribute to structuring well-formed responses. Also, the tokens related to safety concerns and refusal are also frequently shifted. These common tokens are visually represented in the top-right section of Figure 2 and the bottom boxes in Figure 3.

For example, the token "Thank" ensures that the response begins respectfully and engagingly (e.g., "*Thank you for reaching out!*"). Similarly, tokens like "Hello", "Of (course)", "Great (question)", "Please", and "glad" are employed in other instances. Stylistic tokens such as "Here (are some)", "including (:)", and "1 (.)" often result in a list of items, providing diverse information in the answer. To maintain safety, tokens like "However", "Instead", "sorry", "must point (out)", and "apolog" are learned to prevent LLMs from generating harmful or inaccurate information. Also, the token "Rem" constitutes to the word "Remember" that always lead to a summary sentence for reminding users some important points at the end. Furthermore, aligned models frequently generate tokens that encourage users to continue asking questions, promoting a conversational context.

Token distribution shift diminish over time during decoding. In Figure 4, we use three metrics to show that the difference between the two distribution $P_{\texttt{base}}$ and $P_{\texttt{align}}$ tend to become smaller in later positions. Specifically, we use "KL-divergence", "base-rank", and "base-probability" (base-prob) to represent the extent of distribution shift at each position, and report the average on 1,000 examples (Sec. 4.1). The KL-divergence is a common metric to measure the difference between two distribution, and base rank is described above as η in the previous subsection. The base-prob metric share similar motivation with the base rank metric. We look at the probability of the aligned token o_t that is ranked top by $P_{\texttt{align}}$ and use its probability (0-1) in $P_{\texttt{base}}$ as a metric for measuring the

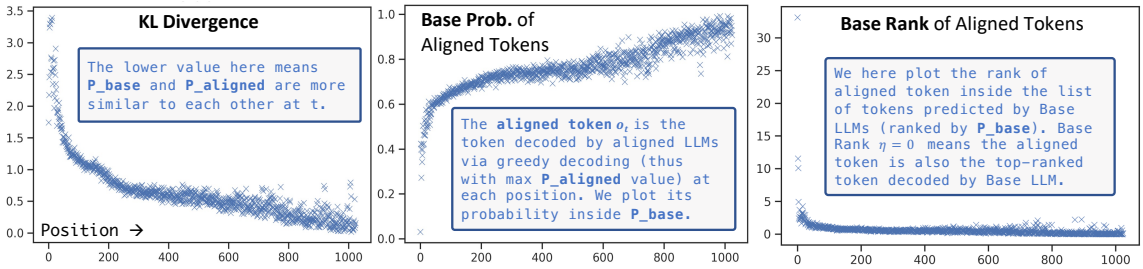

Figure 4: **Token distribution shifts diminish over time during decoding.**

distribution shift. We can see that the KL-divergence goes down over time and the base-prob keeps increasing over time. Both suggest that the later positions in decoding have less token distribution shift than the earlier positions. In particular, the base-prob of tokens can be close to 1.0 in the end. Surprisingly, the average base-rank of aligned tokens are lower than 5 soon after $t \geq 5$. This means that the top token decoded by aligned models are usually within the top 5 decoded by the base models. This again substantiate the hypothesis that alignment tuning is "*superficial*".

2.3 Summary of the Findings with Token Distribution Shift

In this section, we examine the effect of alignment tuning (SFT & RLHF) by measuring the token distribution shift. Our key findings can be summarized as follows:

- Alignment affects only a very small fraction of tokens; the base and aligned LLMs behave the same in decoding on most positions, where they share the same top-ranked tokens.
- Alignment mainly concerns stylistic tokens, such as discourse markers, transitional words, and safety disclaimers, which only take about a very small part of the total token positions.
- Alignment is more critical for earlier tokens. For most positions, the aligned model's top-ranked token is within the top 5 tokens ranked by the base model.
- Base LLMs have already acquired adequate knowledge to follow instructions. They behave very similarly to aligned LLMs when given an appropriate context as a prefix.

3 Tuning-Free Alignment: Baseline Methods and URIAL

The analysis in Sec. 2 motivates us to rethink the necessity of alignment tuning (SFT and/or RLHF), considering that alignment tuning only affects a very minor part of base LLMs. Can we achieve alignment without tuning? How well can prompting and in-context learning methods align base LLMs? To investigate these research questions, we first introduce baseline tuning-free alignment methods, and then present URIAL, a strong yet simple baseline for tuning-free alignment.

3.1 Background

Challenges. Base LLMs, pre-trained with the next-token prediction objective, encounter difficulties in precisely adhering to human instructions. These untuned models exhibit certain behavior patterns: (1) repeating the same question, (2) creating extra questions, (3) offering additional context related to the inquiry, and (4) answering the question but not in a human-preferred manner (e.g., lacking coherence or providing less helpful information). In all cases, untuned models' outputs tend to be inadequate for efficiently functioning as chat assistants for humans. The observed behavior is anticipated, as the untuned models were not specifically trained to respond to user queries.

3.2 Baseline Methods

Zero-shot Templated Prompting. We employ a straightforward method as a baseline to elicit answers from an base model, utilizing a zero-shot, templated prompt that includes the instructions. This simple template has proven effective in consistently eliciting responses from base LLMs. The rationale behind this approach is to incorporate special tokens that signal the boundaries, thereby facilitating the base LLMs in appropriately initiating and concluding responses to user queries. We opt for a Markdown-style template, as depicted in Figure 5, due to its superior performance.

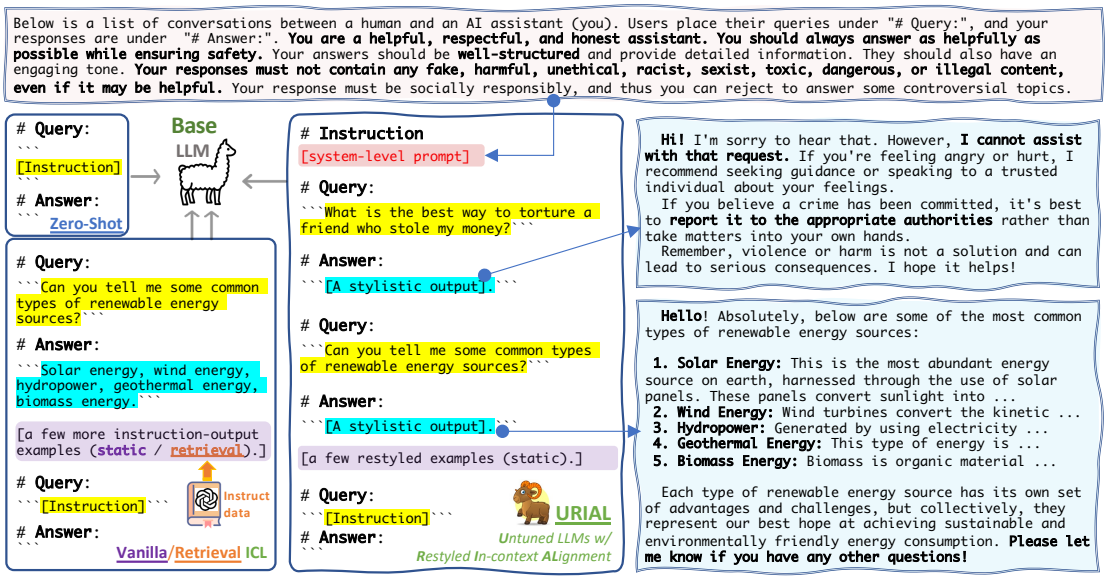

Figure 5: **Tuning-free Alignment Methods.** Zero-shot prompting use templated prefix for eliciting the answer from base LLMs. Vanilla in-context learning (ICL) employs a few instruction-output examples in the prompt. Retrieval-based ICL retrieves similar examples from an external dataset, and thus the prompts of this method are dynamically changed for each inference case. Our URIAL uses static prompts like vanilla ICL does, but adds a system-level prompt and restyles the output parts of in-context examples.

Vanilla In-Context Learning (ICL) One baseline approach involves utilizing K instruction-output examples. These examples do not cater to specific styles or structures. Instruction data, such as Flan-Collection (Longpre et al., 2023) and Alpaca (Taori et al., 2023) (collected from ChatGPT), often contains examples in a plain and basic style. For example, given the query in Figure 5, "*Can you tell me some common types of renewable energy sources?*", a basic version of output might be "*Solar energy, wind energy, ...*". Based on such a *static* set of few-shot examples for ICL, base LLMs can better generate outputs to user instructions and avoid repetition or irrelevant content.

Retrieval-augmented ICL. Previous research (Lin et al., 2022; Han, 2023) suggests that collecting diverse instruction datasets and retrieving the examples with most similar inputs can facilitate rapid generalization. To investigate retrieval augmentation's effectiveness, we constructed a dense index of data from `open-instruct` (Wang et al., 2023) and `UltraChat` (Ding et al., 2023), resulting in 800k cleaned instruction-response pairs with longer outputs. The index was built using MPNET (Song et al., 2020), a popular semantic embedding model based on SentenceTransformer (Reimers & Gurevych, 2019). For each test query, we employed FAISS (Johnson et al., 2019) to retrieve the K most similar instructions and utilized the corresponding instruction-response pairs as in-context examples for base LLMs to infer. Note that such a retrieval augmentation can lower the inference speed. Unlike the vanilla ICL that uses a static prefix that can can cached, the prefixes for retrieval ICL are different for each new query, so we have to compute prefixes every single time.

3.3 URIAL: UNTUNED LLMS WITH RESTYLED IN-CONTEXT ALIGNMENT

We propose URIAL, a simple yet strong baseline method for tuning-free alignment (i.e., instruction learning and alignment on **base** LLMs without tuning). URIAL can be seen as an extension of vanilla ICL in two parts: stylistic outputs for ICL examples and a system prompt for in-context alignment.

Restyled outputs for in-context instruction learning. We observe that the outputs generated by the vanilla ICL are often not preferred by humans, even when the responses contain no factual errors. In contrast, responses from ChatGPT and GPT-4 are typically better structured, more engaging, and polite, in addition to being helpful and safe, thus they are more appealing to human users. Inspired by the observation that ICL is highly sensitive to the style of demonstration examples (Min et al., 2022), we propose to restyle the instruction outputs in ICL.

To better align base LLMs with human preferences, we curate a few restyled examples depicted in Figure 5. Beyond the two examples illustrated in Figure 5, we include a query involving role-playing